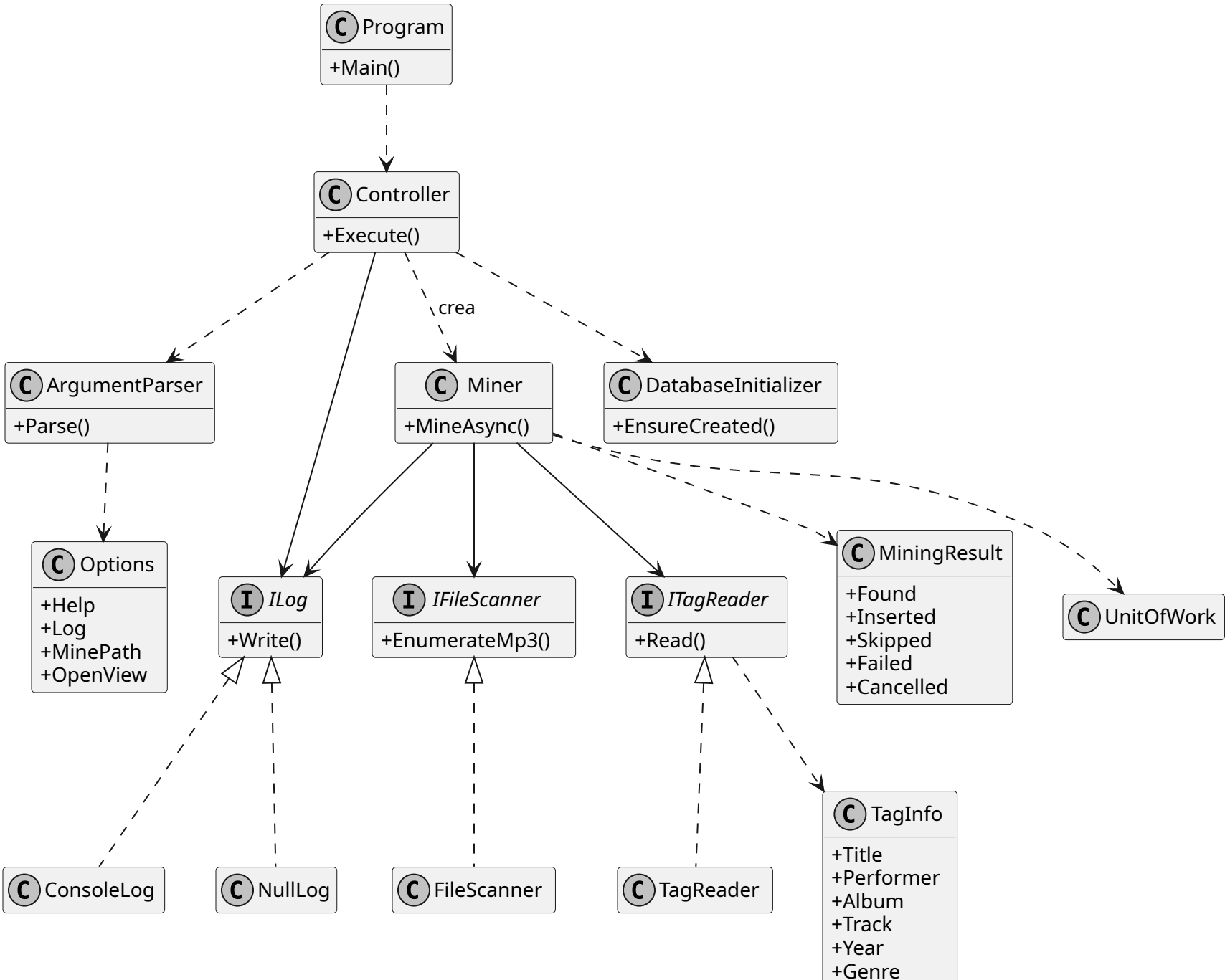Controlador, bitácora y minero